

Deflection of beams

We have already derived the pure bending equation for symmetric, prismatic, homogeneous beams as $\sigma_{xx} = \frac{-E}{\rho} y = -\frac{M_z}{I_{zz}} y$

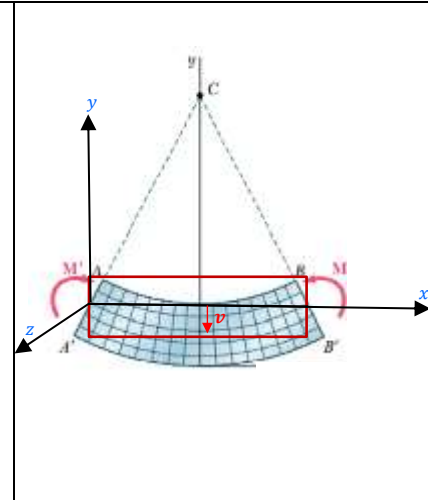
$$\frac{1}{\rho} = \frac{M_z}{EI_{zz}}$$

Let v be the deflection of the beam from x -axis downwards. Then $v = f(x)$ is the equation of the elastic curve. From the elementary calculus, the

equation for curvature is $\frac{1}{\rho} = \frac{\frac{d^2v}{dx^2}}{\left\{1 + \left(\frac{dv}{dx}\right)^2\right\}^{\frac{3}{2}}}$

For small deformations, the deflection curve is shallow. The slope $\frac{dv}{dx} = \tan \theta \cong \theta$ (in radians) is small.

Then $\left(\frac{dv}{dx}\right)^2$ is still smaller. Then $1 + \left(\frac{dv}{dx}\right)^2 \cong 1$.

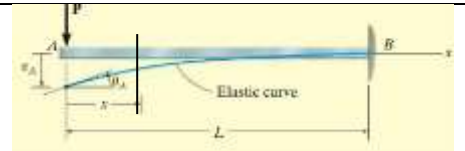


$$\frac{1}{\rho} = \frac{d^2v}{dx^2} = \frac{M_z}{EI_{zz}}$$

The second order differential equation $EI_{zz} \frac{d^2v}{dx^2} = M_z$ is the equation for the elastic curve.

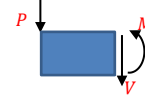
Example 1

Determine the slope and deflection of the cantilever beam loaded at the end.



$$\Sigma M_{1-1} = 0 \ll \gg M + Px = 0 \gg M = -Px$$

$$EI_{zz} \frac{d^2v}{dx^2} = M_z = -Px$$



Integrating with respect to x ,

$$\int EI_{zz} \frac{d^2v}{dx^2} dx = \int -Px dx \ll \gg EI_{zz} \frac{dv}{dx} = -\frac{Px^2}{2} + C_1$$

Integrating again

$$EI_{zz} v = -\frac{Px^3}{6} + C_1 x + C_2$$

Substituting the boundary conditions: At $x = L$, $\frac{dv}{dx} = 0$ and $v = 0$

$$0 = -\frac{PL^2}{2} + C_1 \ll \gg C_1 = \frac{PL^2}{2}$$

$$0 = -\frac{PL^3}{6} + \left(\frac{PL^2}{2}\right)L + C_2 \ll \gg C_2 = -\frac{PL^3}{3}$$

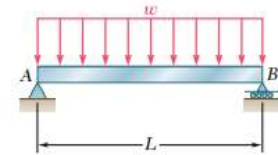
$$EI_{zz} \frac{dv}{dx} = -\frac{Px^2}{2} + \frac{PL^2}{2} \ll \gg EI_{zz} v = -\frac{Px^3}{6} + \left(\frac{PL^2}{2}\right)x - \frac{PL^3}{3}$$

At the free end of the cantilever beam, $x = 0$

$$\frac{dv}{dx} = \frac{PL^2}{2EI_{zz}} \ll \gg v = -\frac{PL^3}{3EI_{zz}}$$

#Example 2

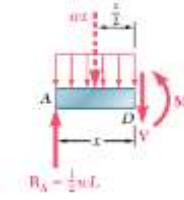
A simply supported beam with uniformly distributed load of intensity w



The total load is wL . The support reactions are $\frac{wL}{2}$.

$$\Sigma M_{1-1} = 0 \lll M - \frac{wL}{2}x + \frac{wx^2}{2} = 0 \ggg M = \frac{wL}{2}x - \frac{wx^2}{2}$$

$$EI_{zz} \frac{d^2v}{dx^2} = M_z = \frac{wL}{2}x - \frac{wx^2}{2}$$



Integrating with respect to x ,

$$\int EI_{zz} \frac{d^2v}{dx^2} dx = \int \left(\frac{wL}{2}x - \frac{wx^2}{2} \right) dx \lll EI_{zz} \frac{dv}{dx} = \frac{wLx^2}{4} - \frac{wx^3}{6} + C_1$$

Integrating again

$$EI_{zz}v = \frac{wLx^3}{12} - \frac{wx^4}{24} + C_1x + C_2$$

Substituting the boundary conditions: At $x = 0$, $v = 0$ and at $x = L$, $v = 0$

$$0 = C_2$$

$$0 = \frac{wL^4}{12} - \frac{wL^4}{24} + C_1L \ggg C_1 = -\frac{wL^3}{24}$$

$$EI_{zz} \frac{dv}{dx} = \frac{wLx^2}{4} - \frac{wx^3}{6} - \frac{wL^3}{24} \lll EI_{zz}v = \frac{wLx^3}{12} - \frac{wx^4}{24} - \frac{wL^3}{24}x$$

When $x = L/2$

$$v = -\frac{5wL^4}{384EI_{zz}} \lll \frac{dv}{dx} = 0$$

Exercise 1

Determine the slope and deflection at the free end of a cantilever of length L and carrying a uniformly distributed load of intensity w N/m.

Exercise 2

Determine the deflection of the beam at the centre of a simply supported beam loaded by a concentrated load P at its centre.

Macaulay's Method

Macaulay's **singularity** functions can be used to determine the deflection of beams, which can be written as

$$\langle x - a \rangle^n = \begin{cases} 0 & \text{for } x < a \\ (x - a)^n & \text{for } x \geq a \end{cases}$$

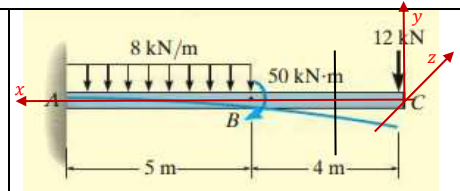
$$n \geq 0$$

The integration of the Macaulay's function follows the same rules as for ordinary functions.

$$\int \langle x - a \rangle^n dx = \frac{\langle x - a \rangle^{n+1}}{n+1} + C$$

#Example 1

Determine the equation of the elastic curve for the beam loaded as shown. Also determine the slope and deflection at the free end of the beam.



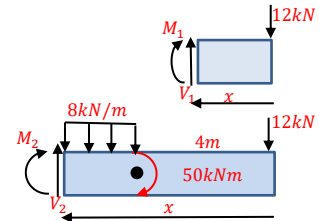
Section 1-1

$$\Sigma M_{1-1} = 0 \ll \gg -M_1 - 12x = 0 \gg M_1 = -12x$$

Section 2-2

$$\Sigma M_{2-2} = 0 \ll \gg -M_2 - 12x - 50 - 8 \frac{(x-4)^2}{2} = 0$$

$$M_2 = -12x - 50 - 4(x-4)^2$$



Now, we can write a single moment equation for the entire beam using the singularity functions

$$M = -12x - 50\langle x - 4 \rangle^0 - 4\langle x - 4 \rangle^2$$

The deflection equation is

$$EI_{zz} \frac{d^2 v}{dx^2} = M_z = -12x - 50\langle x - 4 \rangle^0 - 4\langle x - 4 \rangle^2$$

Integrating with respect to x,

$$\int EI_{zz} \frac{d^2 v}{dx^2} dx = \int \{-12x - 50\langle x - 4 \rangle^0 - 4\langle x - 4 \rangle^2\} dx$$

$$EI_{zz} \frac{dv}{dx} = \frac{-12x^2}{2} + C_1 - \frac{50\langle x - 4 \rangle^1}{1} - \frac{4\langle x - 4 \rangle^3}{3}$$

Integrating again

$$EI_{zz} v = \frac{-6x^3}{3} + C_1 x + C_2 - \frac{50\langle x - 4 \rangle^2}{2} - \frac{4\langle x - 4 \rangle^4}{12}$$

Substituting the boundary conditions: At $x = 9\text{m}$, $\frac{dv}{dx} = 0$ and $v = 0$

$$0 = \frac{-12 \times 9^2}{2} + C_1 - \frac{50(9-4)^1}{1} - \frac{4(9-4)^3}{3} \gg \gg C_1 = 902.7$$

$$0 = \frac{-6 \times 9^3}{3} + 902.7 \times 9 + C_2 - \frac{50(9-4)^2}{2} - \frac{4(9-4)^4}{12} \gg \gg C_2 = -5832$$

The equation for the slope of the elastic curve is

$$EI_{zz} \frac{dv}{dx} = -6x^2 + 902.7 - 50\langle x - 4 \rangle^1 - 1.33\langle x - 4 \rangle^3$$

The equation of the elastic curve is

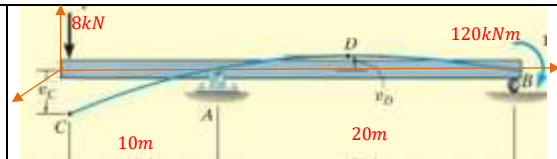
$$EI_{zz} v = -2x^3 + 902.7x - 5832 - 25\langle x - 4 \rangle^2 - 0.33\langle x - 4 \rangle^4$$

At the free end of the cantilever ($x = 0$)

$$\frac{dv}{dx} = \frac{902.7}{EI_{zz}} \ll \gg v = \frac{-5832}{EI_{zz}}$$

#Example 2

Determine the deflections at C and D of the overhanging beam shown.



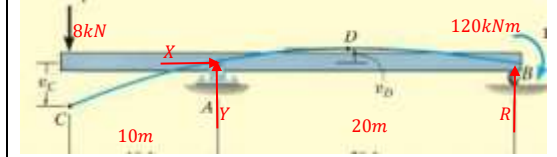
Let us find out the support reactions first.

$$\Sigma F_x = 0 \gg X = 0$$

$$\Sigma M_A = 0 \gg 8 \times 10 + R \times 20 - 120 = 0$$

$$R = 2 \text{ kN}$$

$$\Sigma F_y = 0 \gg Y + R - 8 = 0 \gg Y = 6 \text{ kN}$$



Section 1-1

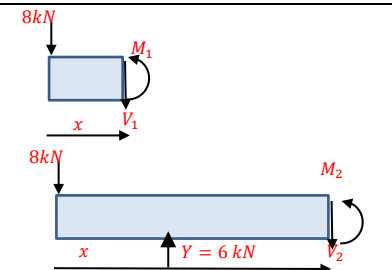
$$\Sigma M_{1-1} = 0 \ll +M_1 + 8x = 0 \gg M_1 = -8x$$

Section 2-2

$$\Sigma M_{2-2} = 0 \ll +M_2 + 8x - 6(x - 10) - 120(x - 30)^0 = 0$$

$$M_2 = -8x + 6(x - 10) + 120(x - 30)^0$$

We know that there is a moment of -120 kNm at $x = 30 \text{ m}$



Hence, we can write a single moment equation for the entire beam using the singularity functions as

$$M = -8x + 6\langle x - 10 \rangle^1 + 120\langle x - 30 \rangle^0$$

$$EI_{zz} \frac{d^2v}{dx^2} = M_z = -8x + 6\langle x - 10 \rangle^1 + 120\langle x - 30 \rangle^0$$

Integrating with respect to x ,

$$EI_{zz} \frac{dv}{dx} = \frac{-8x^2}{2} + C_1 + \frac{6\langle x - 10 \rangle^2}{2} + \frac{120\langle x - 30 \rangle^1}{1}$$

Integrating again

$$EI_{zz} v = \frac{-4x^3}{3} + C_1 x + C_2 + \frac{3\langle x - 10 \rangle^3}{3} + \frac{120\langle x - 30 \rangle^2}{2}$$

Substituting the boundary conditions: At $x = 10 \text{ m}$, $v = 0$ and at $x = 30 \text{ m}$, $v = 0$

$$0 = \frac{-4 \times 10^3}{3} + C_1 \times 10 + C_2 \gg 10C_1 + C_2 = 1333.3$$

$$0 = \frac{-4 \times 30^3}{3} + C_1 \times 30 + C_2 + \frac{3(30 - 10)^3}{3} \gg 30C_1 + C_2 = 28000$$

Solving, we get $C_1 = 1333.3$, $C_2 = -12000$

The equation for the slope of the elastic curve is

$$EI_{zz} \frac{dv}{dx} = -4x^2 + 1333.3 + 3\langle x - 10 \rangle^2 + 120\langle x - 30 \rangle$$

The equation of the elastic curve is

$$EI_{zz} v = -1.33x^3 + 1333.3x - 12000 + \langle x - 10 \rangle^3 + 60\langle x - 30 \rangle^2$$

At the free end of the cantilever ($x = 0$)

$$\frac{dv}{dx} = \frac{1333.3}{EI_{zz}} \ll \gg v_c = \frac{-12000}{EI_{zz}}$$

At D, deflection is maximum. Hence, the slope at D (which is at $x > 10 \text{ m}$) should be zero

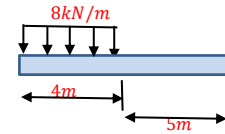
$$0 = -4x^2 + 1333.3 + 3(x - 10)^2 \gg x^2 + 60x - 1633 = 0 \gg x = 20.3 \text{ m}$$

The deflection at D

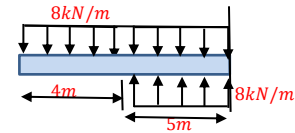
$$v_D = \frac{+5030}{EI_{zz}}$$

#Example 3

Determine the slope and deflection of the cantilever beam shown.



The distributed load acting over 4m is assumed to be acting over the entire 9m downwards and acting over 5m upwards as shown (superposition)



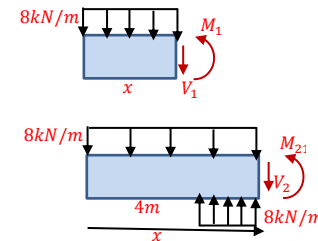
Section 1-1

$$\Sigma M_{1-1} = 0 \ll \gg M_1 + \frac{8x^2}{2} = 0 \gg M_1 = -4x^2$$

Section 2-2

$$\Sigma M_{2-2} = 0 \ll \gg M_2 + \frac{8x^2}{2} - 8 \frac{(x-4)^2}{2} = 0$$

$$M_2 = -4x^2 + 4(x-4)^2$$



we can write a single moment equation for the entire beam using the singularity functions as

$$M = -4x^2 + 4(x-4)^2$$

$$EI_{zz} \frac{d^2v}{dx^2} = M_z = -4x^2 + 4(x-4)^2$$

Integrating with respect to x,

$$EI_{zz} \frac{dv}{dx} = \frac{-4x^3}{3} + C_1 + \frac{4(x-4)^3}{3}$$

Integrating again

$$EI_{zz}v = \frac{-4x^4}{12} + C_1x + C_2 + \frac{4(x-4)^4}{12}$$

Substituting the boundary conditions: At $x = 9m$, $\frac{dv}{dx} = 0$ and $v = 0$

$$0 = \frac{-4 \times 9^3}{3} + C_1 + \frac{4(9-4)^3}{3} \gg \gg C_1 = 805.3$$

$$0 = \frac{-4 \times 9^4}{12} + 805.3 \times 9 + C_2 + \frac{4(9-4)^4}{12} \gg \gg C_2 = -5894$$

The equation for the slope of the elastic curve is

$$EI_{zz} \frac{dv}{dx} = \frac{-4x^3}{3} + 805.3 + \frac{4(x-4)^3}{3}$$

The equation of the elastic curve is

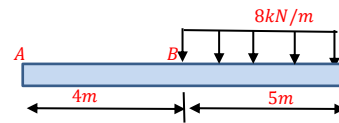
$$EI_{zz}v = \frac{-4x^4}{12} + 805.3x - 5894 + \frac{4(x-4)^4}{12}$$

At the free end of the cantilever ($x = 0$)

$$\frac{dv}{dx} = \frac{805.3}{EI_{zz}} \ll \gg v = \frac{-5894}{EI_{zz}}$$

Exercise 1

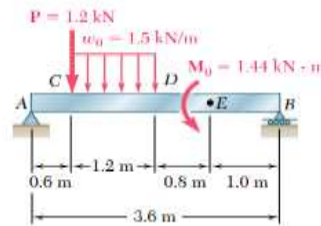
Determine the slope and deflection of the cantilever beam at A and B

**Exercise 2**

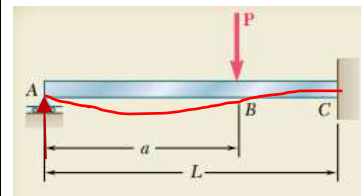
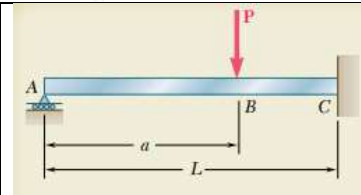
Determine the slope and deflection of the beam directly under the load. What is the maximum deflection and where will it occur?

**Exercise 3**

Determine the maximum deflection of the beam loaded as shown.

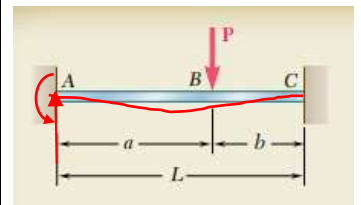
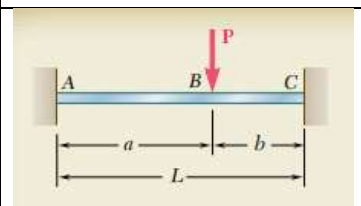
**Exercise 4**

Determine the reaction at A of the propped cantilever beam in terms of $P, a, L, E,$ and I . What is the reaction at A and the deflection under the load when $a = \frac{L}{2}$? (Hint: Replace the roller support at A by the reaction R_A and find the bending moment equation in terms of Macaulay's functions and get the expression for slope and deflection as usual. The expressions will have three unknowns $C_1, C_2,$ and R_A . Substitute the three boundary conditions $(v)_{x=0} = 0, (v)_{x=L} = 0,$ and $\left(\frac{dv}{dx}\right)_{x=L} = 0$ to evaluate the unknowns.)

**Exercise 5**

Determine the reaction at A of the propped cantilever beam in terms of $P, a, L, E,$ and I .

(Hint: Replace the support at A by the reaction R_A and moment M_A . Find the bending moment equation in terms of Macaulay's functions and get the expression for slope and deflection as usual. The expressions will have four unknowns $C_1, C_2, M_A,$ and R_A . Substitute the four boundary conditions $(v)_{x=0} = 0, (v)_{x=L} = 0, \left(\frac{dv}{dx}\right)_{x=0} = 0,$ and $\left(\frac{dv}{dx}\right)_{x=L} = 0$ to evaluate the unknowns.)



Moment-Area Theorems

Consider a beam subjected to some arbitrary loading. The second figure represents the bending moment diagram scaled down by its flexural rigidity (i.e., $\frac{M}{EI}$ diagram).

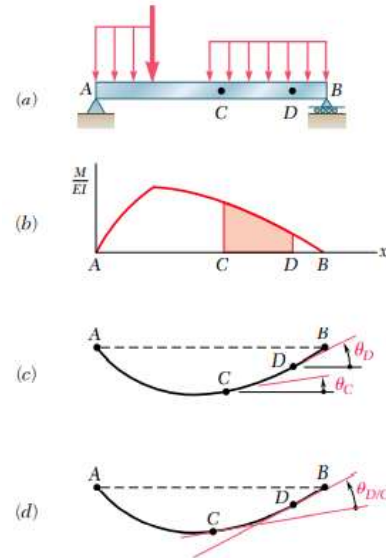
The third figure shows the elastic deflection curve.

We know that the slope, $\theta = \frac{dv}{dx}$

$$\therefore \frac{d^2v}{dx^2} = \frac{M}{EI} \gggg \frac{d\theta}{dx} = \frac{M}{EI} \gggg d\theta = \frac{M}{EI} dx$$

Considering two arbitrary points C and D on the beam and integrating from C to D

$$\int_{\theta_C}^{\theta_D} d\theta = \int_{x_C}^{x_D} \frac{M}{EI} dx$$

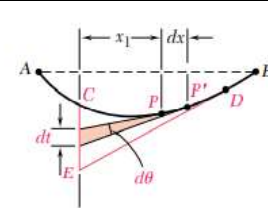


$$\theta_D - \theta_C = \int_{x_C}^{x_D} \frac{M}{EI} dx$$

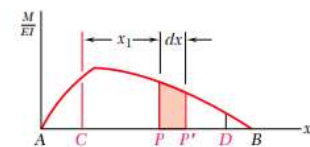
The angle between the tangents at D and C ($\theta_{D/C}$) = Area of the $\frac{M}{EI}$ diagram between C and D. This is known as the **first moment-area theorem**.

Now consider two points P and P' located between C and D. The tangents to the elastic curve at P and P' intersect the vertical through C making an intercept dt. Since dθ is very small, we can write

$$dt = x_1 d\theta \gggg dt = x_1 \frac{M}{EI} dx$$



Now we need to integrate this equation from C to D. As P moves from C to D, the tangent at P sweeps a distance CE along the vertical through C denoted by $t_{D/C}$

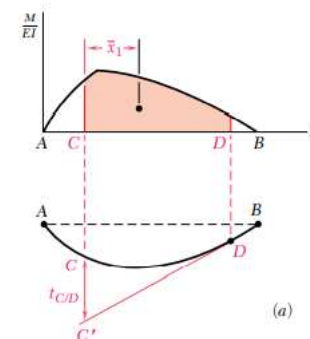


$$t_{D/C} = \int_{x_C}^{x_D} \left(\frac{M}{EI} dx \right) x_1$$

The vertical offset of point C from the tangent at D is equal to the first moment of the area under the $\frac{M}{EI}$ diagram between C and D with respect to the vertical axis through C.

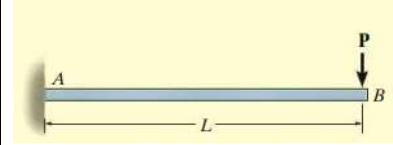
$$t_{D/C} = (\text{Area of } \frac{M}{EI} \text{ diagram})(\bar{x}_1)$$

This is known as the **second moment area theorem**.

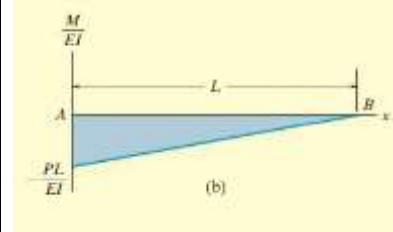


#Example 1

Determine the slope and deflection at B of the cantilever shown using moment-area method. Take EI as constant.



First draw the bending moment diagram and scale it down to $\frac{M}{EI}$



The first moment-area theorem states that

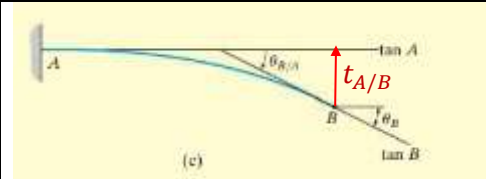
$$\theta_{B/C} = \theta_B - \theta_A = \text{area of } \frac{M}{EI} \text{ diagram between A \& B}$$

$$\theta_B - \theta_A = \frac{1}{2} \left(\frac{-PL}{EI} \right) \times L = -\frac{PL^2}{2EI}$$

Since, $\theta_A = 0$, $\theta_B = -\frac{PL^2}{2EI}$. Negative sign shows that the angle is measured clockwise.

According to the second moment-area theorem

$$t_{A/B} = (\text{Area of } \frac{M}{EI} \text{ diagram b/w A \& B})(\bar{x}_1)$$

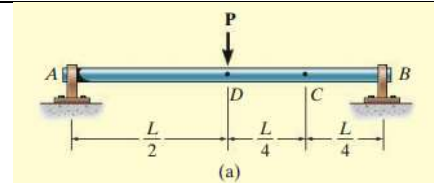


The vertical offset of point B from the tangent at A is equal to the first moment of the area of $\frac{M}{EI}$ diagram between A and B with respect to point B.

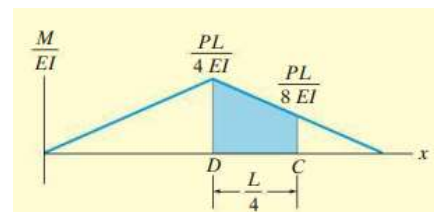
$$t_{A/B} = y_B = \left(-\frac{PL^2}{2EI} \right) \times \frac{2L}{3} = -\frac{PL^3}{3EI}$$

#Example 2

Determine the deflection of point D of the simply supported beam shown. Also calculate the slope and deflection of point C.



First sketch the bending moment diagram and scale it down to $\frac{M}{EI}$ diagram.



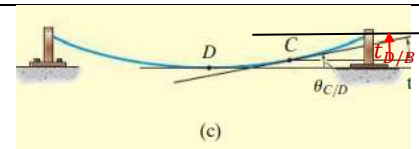
The vertical offset of point B from the tangent at D is equal to the first moment of the area of $\frac{M}{EI}$ diagram between D & B with respect to point B.

$$t_{D/B} = \left(+\frac{1}{2} \times \frac{PL}{4EI} \times \frac{L}{2} \right) \times \frac{2}{3} \left(\frac{L}{2} \right) = +\frac{PL^3}{48EI}$$

$$y_D = -\frac{PL^3}{48EI}$$

$$\theta_{C/D} = \theta_C - \theta_D = \text{area of } \frac{M}{EI} \text{ diagram between C \& D}$$

$$= \frac{1}{2} \times \frac{L}{4} \times \left(\frac{PL}{4EI} + \frac{PL}{8EI} \right) = \frac{3PL^2}{64EI}$$



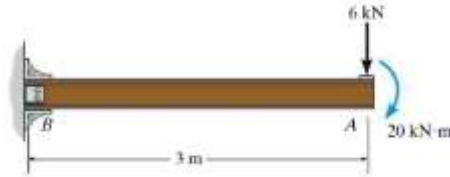
The vertical offset of point C from the tangent at D is equal to the first moment of the area of $\frac{M}{EI}$ diagram between D & C with respect to point C.

$$t_{D/C} = \left(\frac{PL}{8EI} \times \frac{L}{4} \right) \times \left(\frac{L}{8} \right) + \left(\frac{1}{2} \times \frac{L}{4} \times \frac{PL}{8EI} \right) \times \frac{2}{3} \left(\frac{L}{4} \right) = + \frac{5PL^3}{768EI}$$

$$y_c = - \left(\frac{PL^3}{48EI} - \frac{5PL^3}{768EI} \right) = - \frac{11PL^3}{768EI}$$

Exercise 1

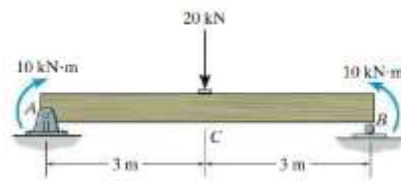
Determine the slope and deflection of point A of the beam with $E = 200 \text{ GPa}$ and $I = 65 \times 10^{-6} \text{ m}^4$

**Exercise 2**

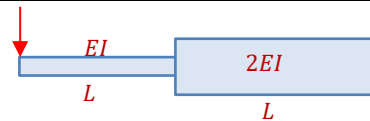
Determine the slope and deflection of the free end of a cantilever beam of length L , flexural rigidity EI carrying a udl of intensity $w \text{ N/m}$ over the entire length.

Exercise 3

Determine the maximum deflection and slope at the ends of the simply supported beam shown

**Exercise 4**

Determine the slope and deflection of the beam at the free end.

**References**

1. Mechanics of Materials, 2/e, J. M. Gere & S. P. Timoshenko, CBS Publishers.
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3. Engineering Mechanics of Solids, 2/e, E. P. Popov, Pearson Education Asia.
4. Mechanics of Materials, 9/e, R. C. Hibbeler, Prentice Hall.